

MULTIMEDIA LAB MANUAL

Computer Science Section

MMV

Course Code: CS204L (Minor)

Subject: Introduction of Multimedia

Duration: 2 Hours

Lab Activity: 02

Target Audience: Undergraduate (UG) Students

Topic: CSS Flexbox & Interactive Card UI

1. Title & Overview

Lab Experiment 03: Designing a Responsive Multimedia Gallery using CSS Flexbox.

Objective: To understand CSS Box Model mechanics, external stylesheet integration, and the Flexbox layout module. Students will construct a dynamic image/content gallery where items automatically wrap, align, and respond to mouse-hover interactions.

2. Problem Statement

You are designing the user interface for your Multimedia Syllabus. Using raw HTML and an external CSS file, create a visually appealing gallery containing five "Content Cards" presenting the various components of Multimedia.

The solution must meet the following structural requirements:

- Use an external stylesheet (style.css) linked to your HTML document.
- Create a main gallery container that utilizes display: flex to align its children in a row and wraps them automatically (flex-wrap: wrap).
- Each card must utilize the CSS Box Model (padding for internal spacing, margin for external spacing, and borders/border-radius).
- Implement a CSS pseudo-class (:hover) to add an interactive transition when the user mouses over a card (e.g., scale the card up or change the background/shadow).
- Change the colors to blue, pink, orange, brown, green respectively for each box.
- Change the Heading as "Components of Multimedia" Using different font style, color, and typeface.
- Add a box in the bottom right corner, half of the size of above boxes. And insert "Created by : Your Name., Roll Number and email ID."

3. Reference

Step 1: Environment Setup

- Create a folder named Lab2_Flexbox_[YourRollNo].
- Open Notepad (or VS Code). Create two files inside this folder: index.html and style.css.
- Notepad Tip: Remember to select "Save as type: All Files (*.*)" so they don't save as .txt.

Step 2: HTML Page Structure

- In index.html, set up the standard HTML5 skeleton.
- In the <head>, link your CSS file using: <link rel="stylesheet" href="style.css">.

- Inside the <body>, create a <div class="gallery-container">.

Step 3: Building the Content Cards

- Inside the gallery container, create three or more cards: <div class="card">.
- Inside each card, add a placeholder colored div or an , an <h3> title, and a <p> description.

Step 4: Writing the CSS (Layout & Box Model)

- In style.css, style the .gallery-container with display: flex;, flex-wrap: wrap;, and justify-content: center;.
- Style the .card class. Give it a specific width (e.g., 100px), padding, border-radius, and a box-shadow to make it look like a physical card.

Step 5: Adding Interactions

- Add a transition: 0.3s; property to the .card class.
- Create a .card:hover rule that modifies the transform: translateY(-10px); or deepens the box-shadow to create a lifting effect.

4. Starter Code Template

Students should start with the following template and edit the sections according to the problem requirements:

File 1: index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>CSS Flexbox Gallery</title>
  <!-- Link to External CSS -->
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <header class="page-header">
    <h1>My Digital Portfolio</h1>
    <p>Showcasing multimedia projects using CSS Flexbox</p>
  </header>

  <!-- The Flexbox Parent -->
  <div class="gallery-container">

    <!-- Card 1 -->
    <div class="card">
      <div class="card-image" style="background-color: #ff7675;"></div>
      <h3>Audio Editing</h3>
      <p>A collection of podcast edits and soundscapes.</p>
    </div>

    <!-- Card 2 -->
    <div class="card">
      <div class="card-image" style="background-color: #74b9ff;"></div>
      <h3>Vector Graphics</h3>
      <p>Logos, icons, and scalable vector illustrations.</p>
    </div>

    <!-- Card 3 (STUDENT TASK: Add a 3rd and 4th card) -->
    <!-- TODO: Add more cards here -->

  </div>
```

```
</body>
</html>
```

File 2: style.css

```
/* Global Styles */
body {
  font-family: Arial, sans-serif;
  background-color: #f4f6f9;
  margin: 0;
  padding: 20px;
}
.page-header {
  text-align: center;
  margin-bottom: 40px;
}

/* Flexbox Gallery Container */
.gallery-container {
  display: flex;
  /* TODO: Add flex-wrap property to allow multiple rows */

  /* TODO: Add justify-content property to center the cards */

  gap: 25px; /* Spacing between flex items */
}

/* Individual Content Card */
.card {
  background-color: #ffffff;
  width: 280px;
  border-radius: 10px;
  padding: 15px;
  box-shadow: 0 4px 6px rgba(0,0,0,0.1);

  /* Smooth transition for hover effects */
  transition: transform 0.3s ease, box-shadow 0.3s ease;
}

/* Image Placeholder inside Card */
.card-image {
  width: 100%;
  height: 160px;
  border-radius: 6px;
  margin-bottom: 15px;
}

/* Hover Interaction */
.card:hover {
  /* TODO: Add a transform property (e.g., translateY) to lift the card */

  /* TODO: Increase the box-shadow to make it look elevated */
  cursor: pointer;
}
```